MA2031-Hints to solve problems of Assignment II

1. Prove atleast one of the conditions for an innerproduct is not satisfied for all the subdivisions

2. Let $f_A(x, y) = y^t Ax$ be an innerproduct on R^2, Assume $x = (x_1, x_2)^t, y = (y_1, y_2)^t$, use the positive definiteness of innerproduct for appropriately chosen vectors x and y so that $a_{11} > 0$, similarly prove a_{22} is positive.

3. Find the orthonormal basis $\mathbb{B}' = \{v^1, v^2, v^3 \cdots, v^n\}$ corresponding to the basis $\mathbb{B} = \{u^1, u^2, u^3 \cdots, u^n\}$. We observe that it is enough to prove $< y, v^i >= 0, \forall i$, Use the fact that $v^i = \sum\limits_{k=1}^{i} \alpha_i u^i$ to prove what is required.

4. (a) Use the fact $< x, x >=|| x ||^2$
   (b) Use the fact $< x, x >=|| x ||^2$ and use the equality of Cauchy schwarz inequality to prove dependence

5. Trivial

6. Use the fact $< x, x >=|| x ||^2$

7. Use Cauchy Schwarz inequality

8. Write the given set as intersection of two planes passing through origin, show that vector addition and scalar multiplication is closed and find the matrix A such that the given space is the solution space of $Ax = 0$, reduce A to rref to find a basis

9. Trivial

10. Trivial

11. Use the fact that $< x, x >=|| x ||^2$, and also use the linear property of innerproduct

12. Trivial from the pythagoras theorem

13. (a) If $x \in V^{\perp}$, then $< x, x >= 0$
    (b) Trivial
    (c) Verify closure property of vector addition and scalar multiplication
    (d) Trivial

14. (a) Start with an orthonormal basis$\{v^1, v^2, \cdots, v^k\}$ for W and extend it so that it is an orthonormal basis $\{v^1, v^2, \cdots, v^k, \cdots, v^n\}$for V,

15. Similar to the problem in sheet 2
16. Apply Gram Schmidt process
17. Write the given space as the solution of a homogeneous linear system and then reduce it to rref to find basis
18. Similar to the previous problem
19. Apply Gram Schmidt to the given vectors
20. Apply Gram Schmidt process
21. (a) Let $u = \alpha v_1 + \beta v_2$, where $v_1 = (3,1,2)^t, v_2 = (1,0,1)^t$, find α and β so that $\| v - u \|^2$ is minimum using calculus or use the property that $< u - v, v_1 >= 0, < u - v_2 >= 0$ to find α, β. Also think of a general formula to find (the) best approximation
    (b) Obtain a basis for the space U and proceed as we did in the previous problem
    (c) Observe that v is in the space U
    (d) Use the idea in (a)
    (e) (a) Apply Gram Schmidt for the columns of the matrix A and use the orthonormality of the columns to find the factorization
        (b) Similar as the above